PHYS 7501:
Particles and Nuclei I

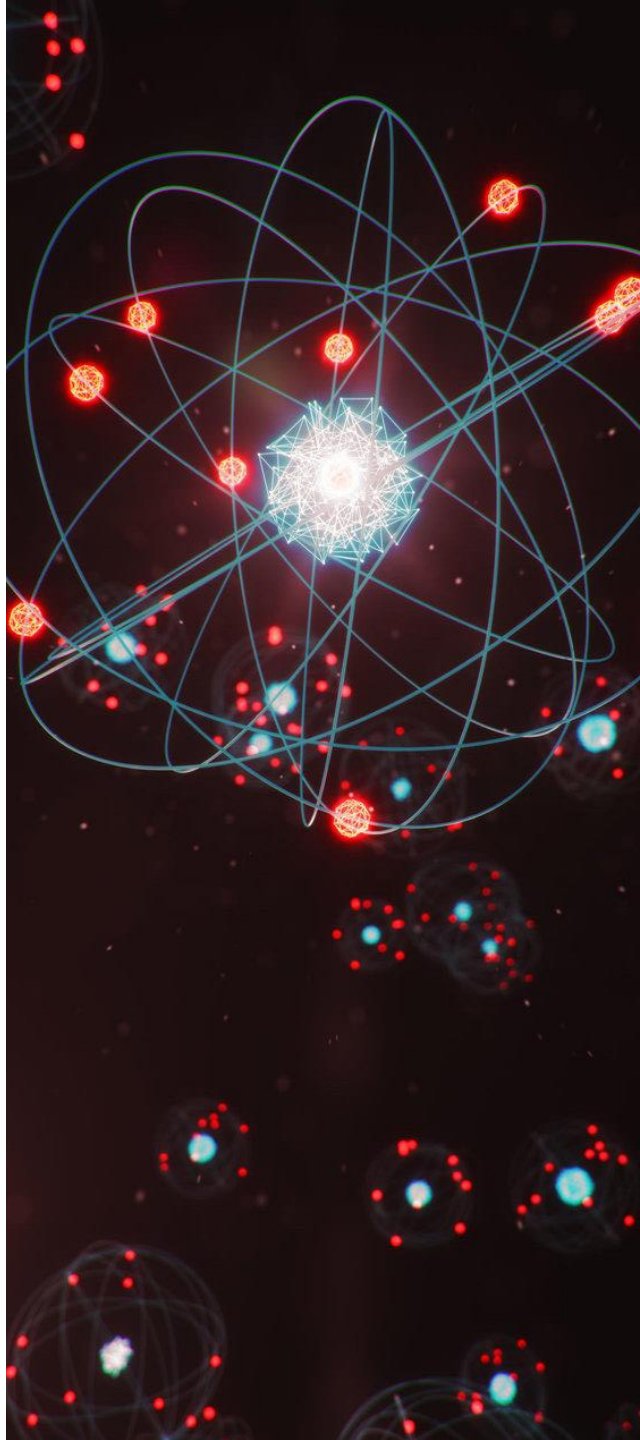
Tues./Thurs. 2 – 3:20PM
Clippinger 105

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NUCLEAR REACTIONS

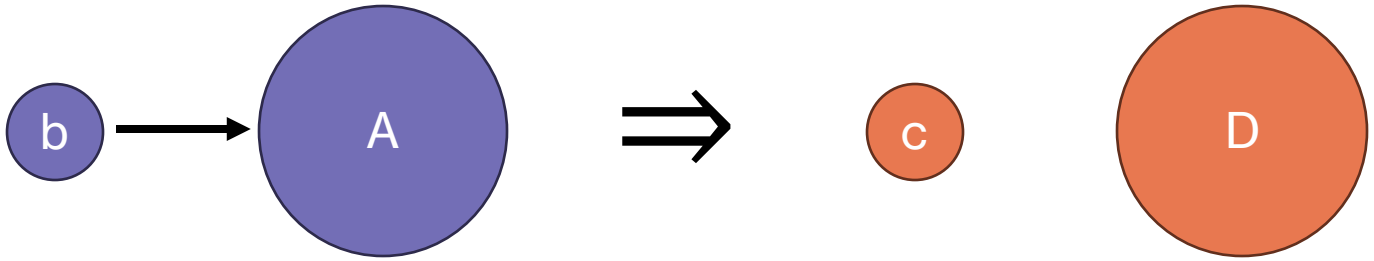
Modern Nuclear Chemistry, Chapter 10

Monday	Tuesday	Wednesday	Thursday	Friday
10/13	10/14	10/15	10/16	10/17
	Nuclear Reactions HW#3 Due		**Scattering**	
10/20	10/21	10/22	10/23	10/24
	Direct Reactions		**Resonant Reactions**	
10/27	10/28	10/29	10/30	10/31
	Statistical Reactions		**Other Reaction Types**	

HW#4 will be due on Nov. 6th

What are Nuclear Reactions?

- A nuclear reaction consists of the interaction of two or more nuclei or nucleons that results in some final product
- Initial participants are called **reactants**, and the final participants are called **products**
- Notation is commonly written as **A + b → c + D** is written as **A(b,c)D**, where **M(b) < M(A)** and **M(c) < M(D)**



- This is called *normal kinematics* where a light beam impinges a heavy target
- Multiple channels can be available for a single set of **reactants**

Classify these Reactions

- ✓ $^{12}C(\alpha,\gamma)^{16}O$
- ✓ $^{12}C(\alpha,n)^{15}O$
- ✓ $^{180}Hf(t,p)^{182}Hf$
- ✓ $^{59}Fe(n,\gamma)^{60}Fe$

Reactants	Products
^{12}C, α	γ, ^{16}O
^{12}C, α	n, ^{15}O
^{180}Hf, t	p, ^{182}Hf
^{59}Fe, n	γ, ^{60}Fe

Notice that $^{12}C+\alpha$ yielded multiple exit channels

Reaction Types

- Reactions are categorized by the **participants** and by the **mechanism**
- **Participants**:
  - Particle comes in and only a **γ** comes out: *radiative capture*
  - **γ** comes in and one or a few particles comes out: *photodisintegration*
  - A multi-nucleon beam transfers a nucleon (or nucleons) to/from the target: *transfer*
  - A projectile exits as an ejectile, taking one or more nucleons with it: *pick-up*
  - A projectile exits as an ejectile, losing one or more nucleons in the process: *knock-out*
  - The projectile and target aren't transmuted and exit in their ground states: *elastic scattering*
  - The projectile and target aren't transmuted and one (or both) is excited: *inelastic scattering*
- **Mechanism**:
  - Few nucleons take part in the reaction (e.g. transfer): *direct*
  - The projectile and target briefly fuse, sharing energy amongst all nucleons: *statistical or compound*
  - The projectile and target briefly fuse, forming a loosely bound state: *resonant*

Examples of Reaction Types

Radiative Capture
(n, γ)
(p, γ)

Photo-disintegration
(γ, n)
(γ, p)

Transfer
(d, p)
(p, t)

Pick-up
(p, d)
(d, t)

Knock-out
$(p, 2p)$

Elastic scattering
(p, p)

Inelastic scattering
(p, p')

Other Types:
Fragmentation
Fusion-Evaporation
Heavy-Ion Collisions

Energetics: *T(P,x)R*

- Projectile **P** + Target **T** → Emitted particle(s) **x** + Residual nucleus **R** + Energy (**Q**), or otherwise written as $\boldsymbol{T(P,x)R}$
- Neglecting election binding energies, the energy can be written as

$$\mathbf{m_P c^2 + T_P + m_T c^2 + T_T = m_R c^2 + T_R + m_x c^2 + T_x}$$

- In general, $\boldsymbol{T_T = 0}$ because the target is stationary.
- Then, the Q-value is defined as

$$\boldsymbol{Q = [m_P + m_T - m_R - m_x]c^2 = T_R + T_x - T_P}$$

- If $Q > 0$, the reaction is exoergic and if $Q < 0$, the reaction is endoergic
- So, a reaction can only be energetically possible if $\mathbf{Q + T_P > 0}$
- If the mass excesses are known, we can write the Q-value as

$$\boldsymbol{Q = \Delta(Projectile) + \Delta(Target) - \sum \Delta(Products)}$$

Energetics: *T(P,x)R*

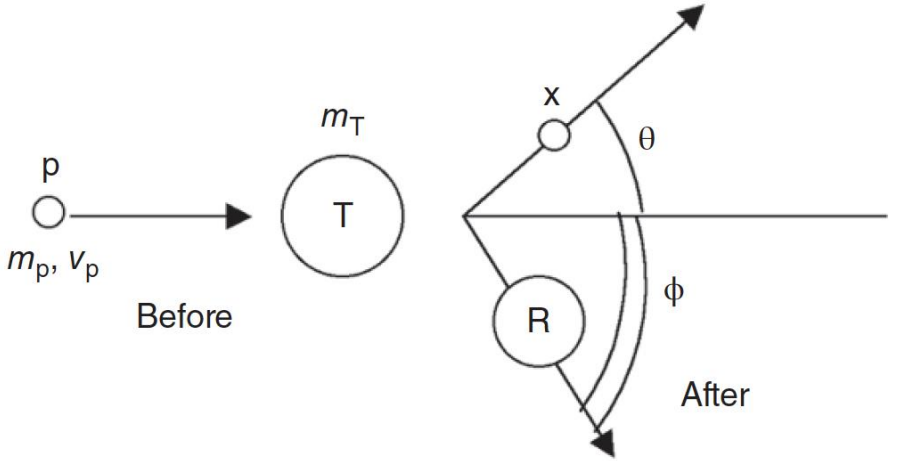
- Momentum conservation in x and y yields:

$$x:\ m_P v_P = m_x v_x \cos\theta + m_R v_R \cos\phi$$

$$y:\ 0 = -m_x v_x \sin\theta + m_R v_R \sin\phi$$

- Recall that momentum is $p = mv = \sqrt{2m_T E_T}$

- Then, substituting into our expressions for x and y then adding the equations:



A typical two-body collision in the laboratory frame

$$m_P T_P - 2\sqrt{m_P T_P m_x T_x}\cos\theta + m_x T_x = m_R T_R$$

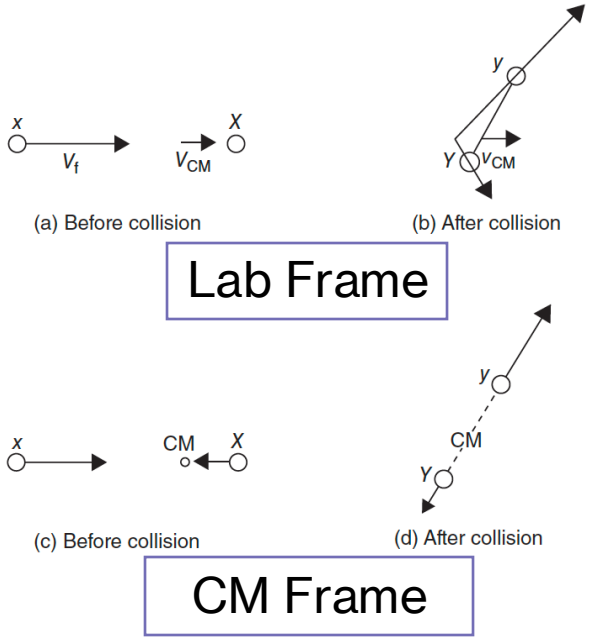
- We know from earlier that $Q = T_R + T_x - T_P$, then we get *the Q-value equation*

$$Q = T_x\left(1 - \frac{m_x}{m_R}\right) - T_P\left(1 - \frac{m_P}{m_R}\right) - \frac{2}{m_R}\sqrt{m_P T_P m_x T_x}\cos\theta$$

- This expression tells us that the Q-value can be determined using kinetic energy and emission angle for the ejectile only

Energy in the Center of Mass Frame

- It is often more convenient to use the center-of-mass (CM) frame
- In the CM frame, the <u>total momentum of the particles is zero</u> before and after the collision
- The kinetic energy of the CM: $\boldsymbol{T_{CM} = \frac{1}{2}(m_P + m_T)v_{CM}^2}$
- For the kinetic energy conversion between the lab and CM:



$$\boldsymbol{T_{CM} = T_{lab}\left[\frac{m_T}{m_P + m_T}\right]}$$

You can also substitute in A_T, A_P here

- So, the CM energy is always lower than the laboratory energy, scaled by choice of beam and target
- Often expressing laboratory energy in MeV/u is more useful
- Example: $^{96}\mathrm{Zr}(\alpha, \mathrm{n})$ at $E_{CM} = 7.68\ MeV$
  - For an α beam on a ^{96}Zr target: $E_{lab} = \frac{96+4}{96} 7.68\ \mathrm{MeV} = 8\ \mathrm{MeV} = \frac{8\ \mathrm{MeV}}{4\mathrm{u}} = 2\ \mathrm{MeV/u}$
  - For a ^{96}Zr beam on an α target: $E_{lab} = \frac{96+4}{4} 7.68\ \mathrm{MeV} = 192\ \mathrm{MeV} = \frac{192\ \mathrm{MeV}}{96\mathrm{u}} = 2\ \mathrm{MeV/u}$

Reaction Thresholds

- If a beam impinges on a target and the reaction is endoergic (endothermic), what should the beam energy be?
  - The Q-value?
  - Actually this would not be enough energy!
- There is a cost from the extra energy carried away from the center-of-mass system
- Based on the expression $\boldsymbol{T_{CM} = T_{lab}\left[\frac{m_T}{m_P+m_T}\right]}$, the lab energy must be $\boldsymbol{T_P \geq -Q\frac{m_P+m_T}{m_T}}$
- This is the minimum kinetic energy to make the reaction proceed, called the *threshold energy*

Thresholds Group Work

What is the threshold energy for ^{14}N(α,p)?

$$^{4}He + ^{14}N \rightarrow p + ^{17}O + \mathrm{Q}$$

First, we need to determine Q:

$$Q = \Delta(\alpha) + \Delta(^{14}N) - \Delta(p) - \Delta(^{17}O)$$

$$Q = 2.425 + 2.863 - 7.289 - (-0.809) = -1.19 MeV$$

Then, the threshold energy is

$$T_{\alpha} \geq -(-1.19\ MeV)\frac{4+14}{14} = 1.53\ MeV$$

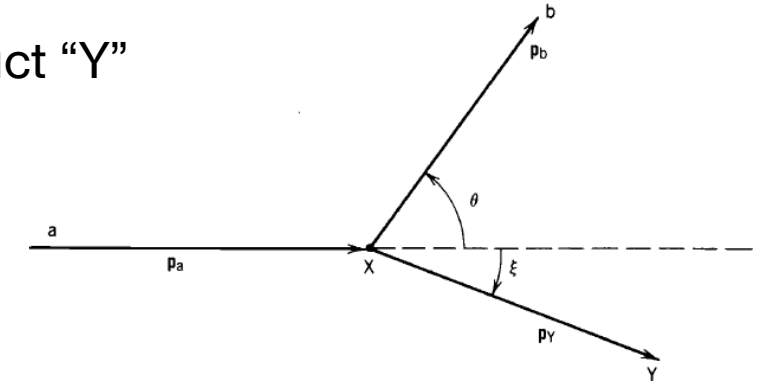
Kinematics

- Using the Q-value equation, $\boldsymbol{Q = T_x\left(1-\frac{m_x}{m_R}\right) - T_P\left(1-\frac{m_P}{m_R}\right) - \frac{2}{m_R}\sqrt{m_P T_P m_x T_x}\cos\theta}$, and conservation of momentum, we can determine the relationship between our reach products outgoing energy and angle

- Call the projectile “a”, target “X”, light product “b”, and heavy product “Y”

$$\sqrt{KE_b} = \frac{\sqrt{m_a m_b KE_a}\cos(\theta) \pm \sqrt{m_a m_b KE_a \cos^2(\theta) + (m_Y + m_b)[m_Y Q + (m_Y - m_a)KE_a]}}{m_Y + m_b}$$

K.S. Krane, Introductory Nuclear Physics (1987)

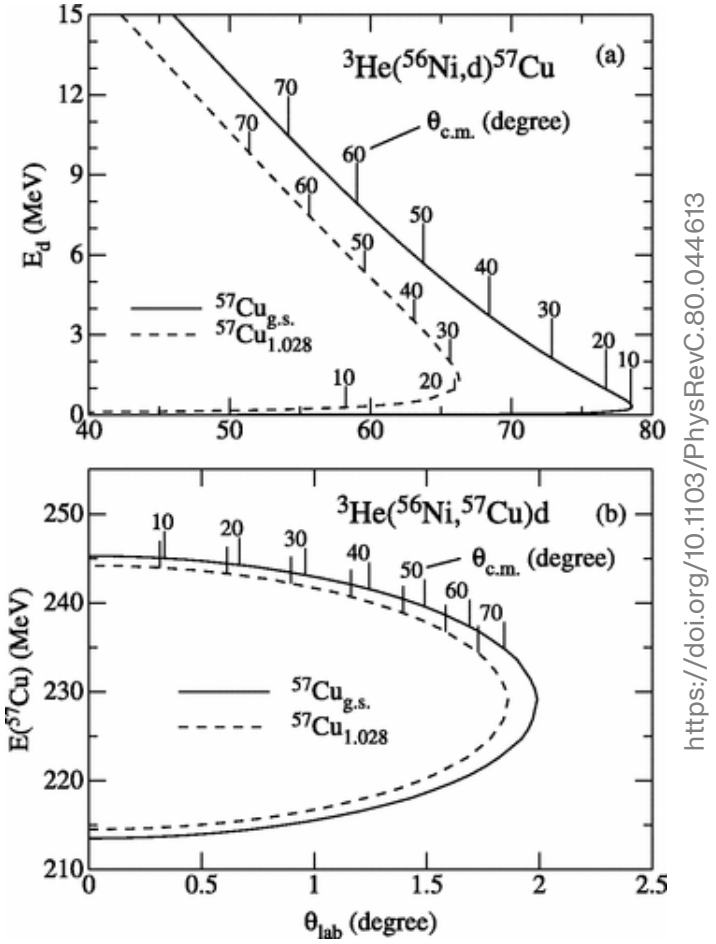


- This is pretty complicated, so most people use calculators that have been developed for two-body kinematics in the laboratory and CM frames

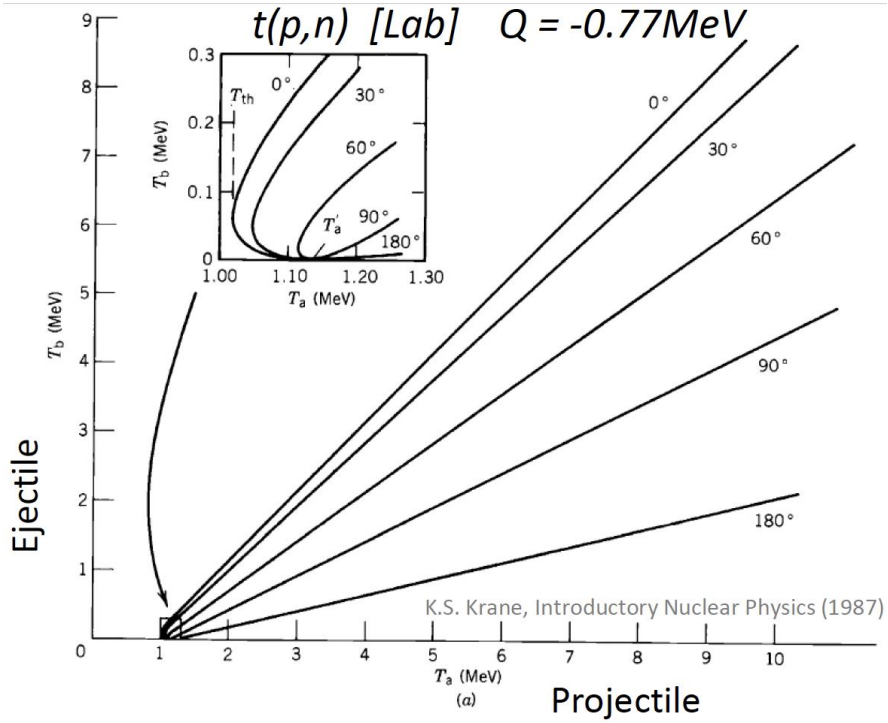
- Common examples: Catkin (Excel Spreadsheet), LISE++ (application from NSCL/FRIB), or SkiSickness (online tool)

Kinematics

- For a given projectile energy, usually only one ejectile energy corresponds to a given angle
- Ejectiles are emitted at 0° for projectiles at the threshold energy



https://doi.org/10.1103/PhysRevC.80.044613



K.S. Krane, Introductory Nuclear Physics (1987)

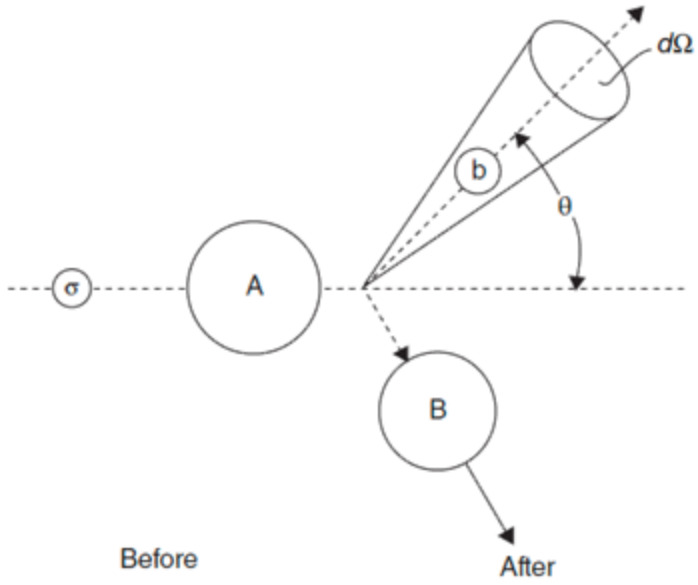
Reaction Likelihood: Cross Section

- To quantify the probability of a nuclear interaction occurring, we use the cross section
- The number of reactions that occur $\boldsymbol{N}$ when we impinge an ion beam of intensity $\mathbf{I}$ (in units of particles/second) for some time $\mathbf{t}$ on a target with areal density $\boldsymbol{n_T}$ (in units of atoms/cm^2) is scaled by a sort of reaction probability σ: (for very thin targets:) $\boldsymbol{N = Itn_T\sigma}$
- For the units to work, σ must have units of area [cm^2] so we call this area the cross section
- You can think of it as the overlap between the wave-like projectile and target, though it can be very different from the classical value from a physical overlap
- In practice, you also need to fold in the detection efficiency (ε) so that $\boldsymbol{\sigma = \frac{N_{detected}}{Itn_T\varepsilon}}$
- The efficiency need not be uniform (and in most cases is not), so we have to add additional considerations

Reaction Likelihood: Cross Section

- Consider the cross section for particles detected at some angle $\boldsymbol{\theta}$ with some solid angle $\boldsymbol{\Omega}$

$$\boldsymbol{\Omega} = \frac{\boldsymbol{Detector\ Area}}{(\boldsymbol{Target\ -Detector\ Distance})^2}$$



Then, we can write

$$\boldsymbol{\sigma} = \frac{\boldsymbol{N_d}}{\boldsymbol{I_b t_{meas} n_T \varepsilon_{det}(\theta) \Omega_{det}(\theta)}}$$

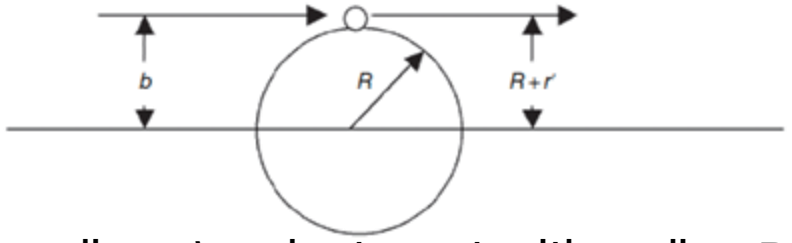
Cross Section Units

- To estimate the scale of cross sections we're going to deal with, consider the area an incoming projectile would see for a typical nucleus
- Take A = 100, for example
  - $R = 1.2A^{1/3} \approx 5.5fm$
  - $\pi R \approx 100fm^2$
  - This is large, so we define a new unit called a **barn**: $1\text{b} = 100fm^2 = 10^{-24}\text{cm}^2$
- Some useful conversions:
  - $1b = 10^{-24}cm^2$
  - $1mb = 10^{-27}cm^2$
  - $1\mu b = 10^{-30}cm^2$
  - $1nb = 10^{-33}cm^2$

In general, cross sections can span many orders of magnitude.

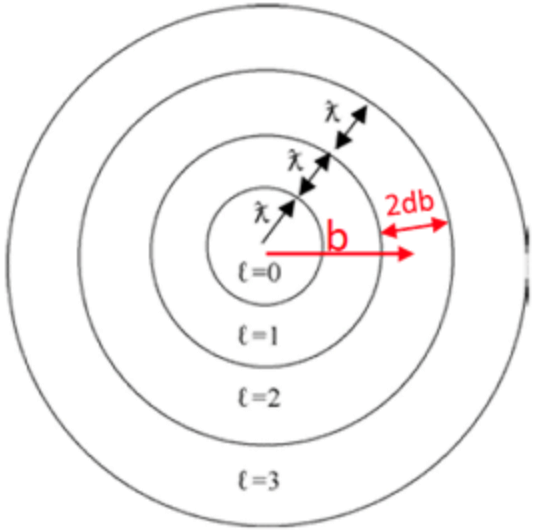
Some neutron-capture cross sections are $10^6 b$, while discovery of new superheavy elements have ~1pb cross sections.

Cross Sections from a Semi-Classical View



- Consider a grazing collision between a projectile with radius $\boldsymbol{r'}$ and a target with radius $\boldsymbol{R}$, where the distance between them is the impact parameter $\boldsymbol{b = R + r'}$
- Using classical mechanics, we can write the orbital angular momentum as

$$\ell = \vec{r} \times \vec{p} = (R + r')p = bp$$



- Since our projectile is a wave, $\boldsymbol{p = \frac{2\pi\hbar}{\lambda}}$, which makes $\ell\hbar = \boldsymbol{2\pi b\frac{\hbar}{\lambda}}$, making $\boldsymbol{b = \frac{\ell}{2\pi\lambda}}$
- Collisions at a given impact parameter $\boldsymbol{b \pm db}$ correspond to a given ℓ
- The geometric cross section for one "slice" is

$$\sigma_\ell = \sigma_{b\pm db} = \sigma_{b+db} - \sigma_{b-db} = \pi(b + db)^2 - \pi(b - db)^2$$

$$= \pi(\ell + 1)^2\left(\frac{\lambda}{2\pi}\right)^2 - \pi(\ell)^2\left(\frac{\lambda}{2\pi}\right)^2 = \pi\left(\frac{\lambda}{2\pi}\right)^2(2\ell + 1)$$

- Including quantum mechanics, the probability of tunneling through a barrier corresponding to ℓ is the transmission coefficient T_ℓ $\sigma_{total} = \left(\frac{\lambda}{2\pi}\right)^2 \sum_\ell (2\ell + 1)T_\ell$ where $0 \leq T_\ell \leq 1$

Cross Sections from a Semi-Classical View: High-Energy Collisions

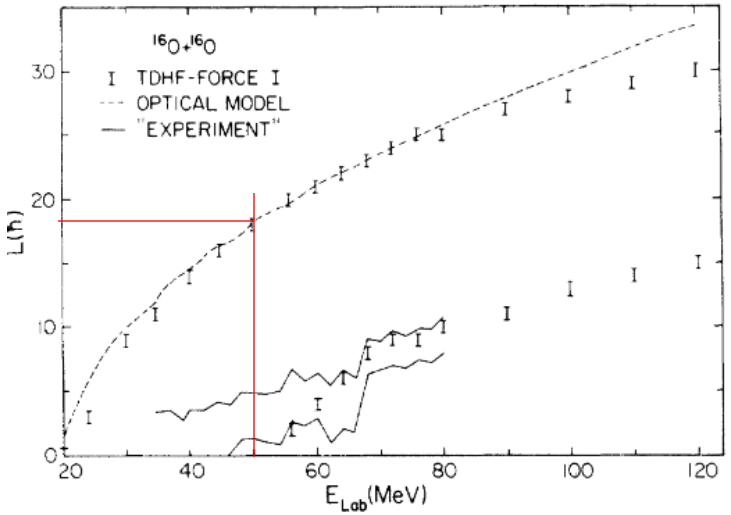
- $\boldsymbol{\sigma_{total}} = \left(\frac{\boldsymbol{\lambda}}{\boldsymbol{2\pi}}\right)^{\boldsymbol{2}} \sum_{\ell}(\boldsymbol{2\ell + 1})\boldsymbol{T}_{\ell}$ where $0 \leq \mathrm{T}_{\ell} \leq 1$

The high-energy limiting case for T_{ℓ}:

- For relatively high-energy (e.g. 100 MeV heavy-ion beams), $\mathrm{T}_{\ell} = \begin{matrix} 1 \text{ for } \ell < \ell_{\max} \\ 0 \text{for } \ell > \ell_{\max} \end{matrix}$

This is the "sharp cutoff limit"

- $\ell_{\max}$ is estimated by assuming almost no beam-target overlap: $\ell_{\max} = \frac{2\pi(\mathrm{R}+\mathrm{r}')}{\lambda}$
- Example: 50 MeV ^{16}O on an ^{16}O target:
- $E_{CM} = 50\frac{16}{16+16} = 25\ MeV$
- $\ell_{\max} = \frac{2\pi(1.2\times16^{1/3}+1.2\times16^{1/3})\mathrm{fm}}{\lambda}\sqrt{2\left(\frac{16\times16}{16+16}\times 931.5\ \mathrm{MeV/u}\right)\times 25\ \mathrm{MeV}} = 19$



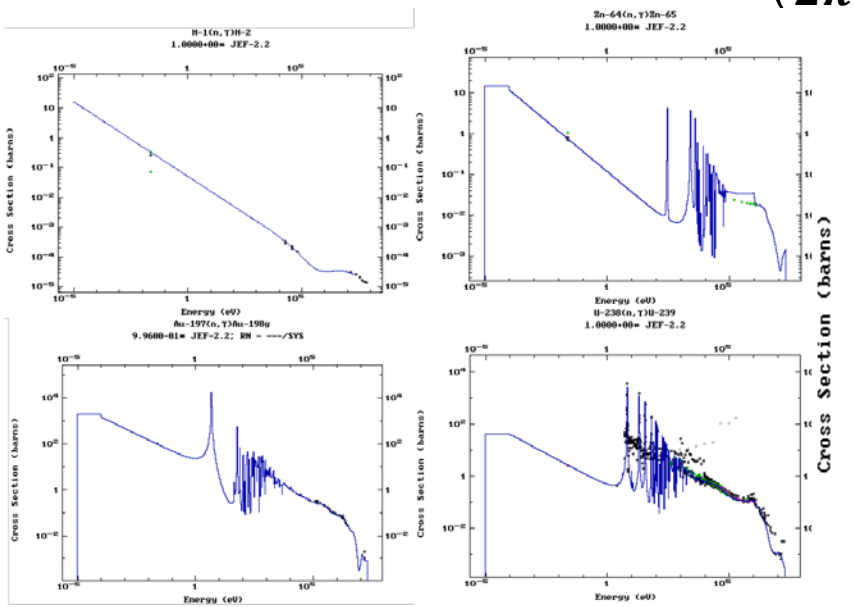
Cross Sections from a Semi-Classical View: Low-Energy Collisions

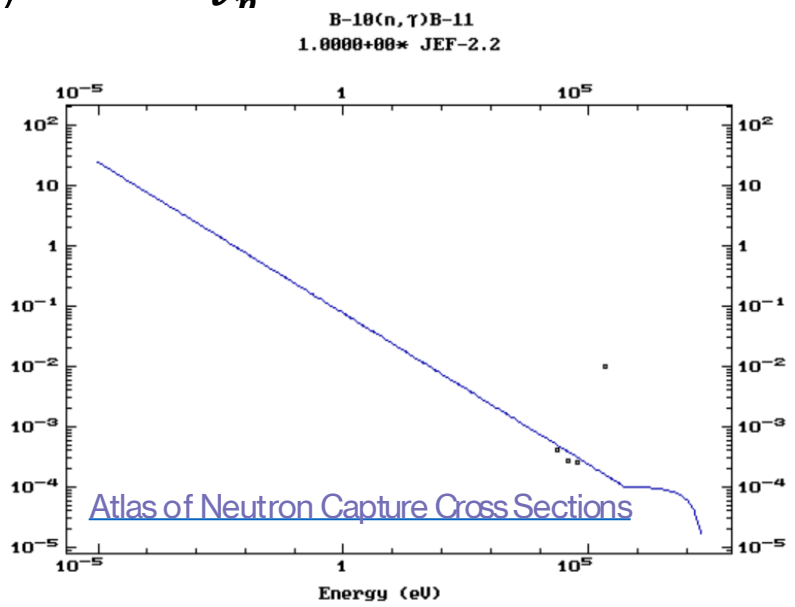
- $\boldsymbol{\sigma_{total}} = \left(\frac{\boldsymbol{\lambda}}{\boldsymbol{2\pi}}\right)^{\boldsymbol{2}} \sum_{\ell}(\boldsymbol{2\ell + 1})\boldsymbol{T}_{\ell}$ where $0 \leq \mathrm{T}_{\ell} \leq 1$
- The low-energy limiting case for T_{ℓ}:
- For relatively low-energy (e.g. < keV beams), $\mathrm{T}_{\ell} \propto \begin{matrix} \sqrt{\mathrm{T_P}} \text{ for } \ell = 0 \\ 0 \text{ for } \ell > 0 \end{matrix}$

This is because lower velocities mean the neutron will spend more time in proximity of the target, and for $\ell > 0$, there isn't enough energy to tunnel

- Consider a neutron at low energy,

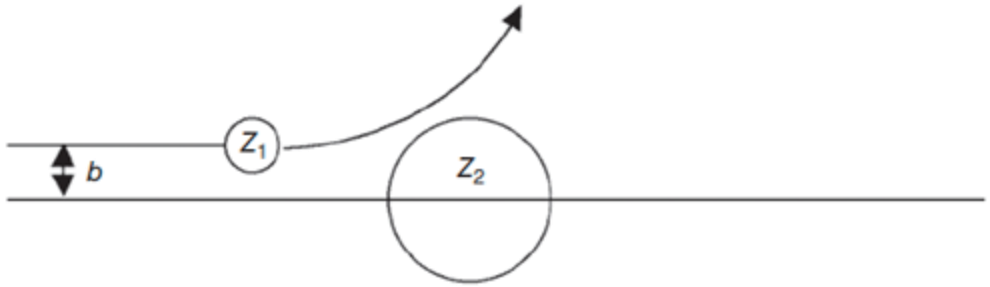
$$\boldsymbol{\sigma_{total}} \propto \left(\frac{\boldsymbol{\lambda}}{\boldsymbol{2\pi}}\right)^{\boldsymbol{2}} \sqrt{\boldsymbol{T_n}} \propto \frac{\boldsymbol{1}}{\boldsymbol{v_n}}$$





Cross Sections from a Semi-Classical View: With Coulomb

- Now, what about Coulomb?
- For a charged-particle, there is a Coulomb barrier between the target and projectile



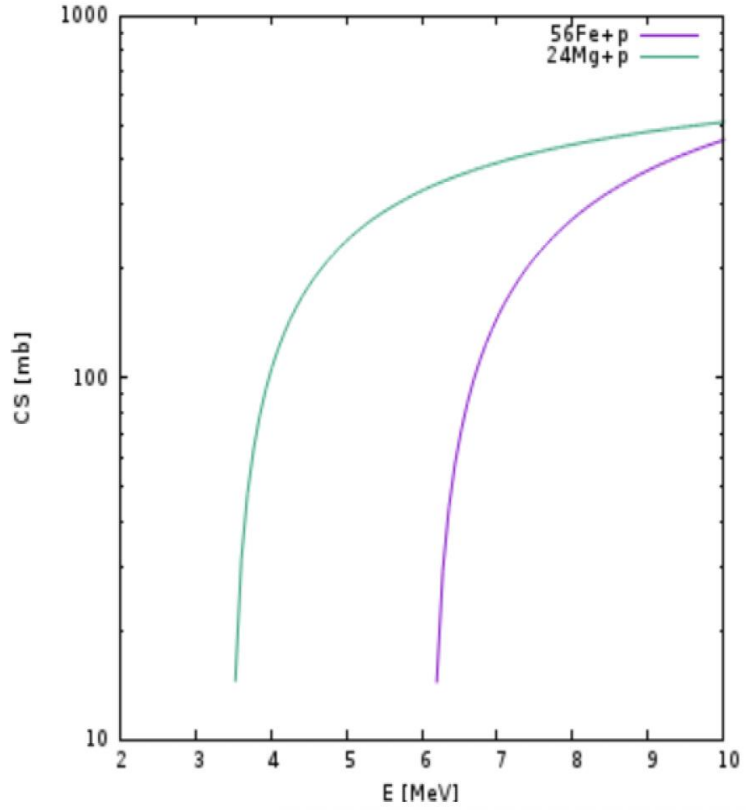
$$V_C = \frac{Z_p Z_t e^2}{R} = \frac{Z_p Z_t}{r_0\left(A_p^{1/3}+A_t^{1/3}\right)}\frac{e^2}{\hbar c}\hbar c \approx 1.44\frac{Z_p Z_t}{r_0\left(A_p^{1/3}+A_t^{1/3}\right)}MeV$$

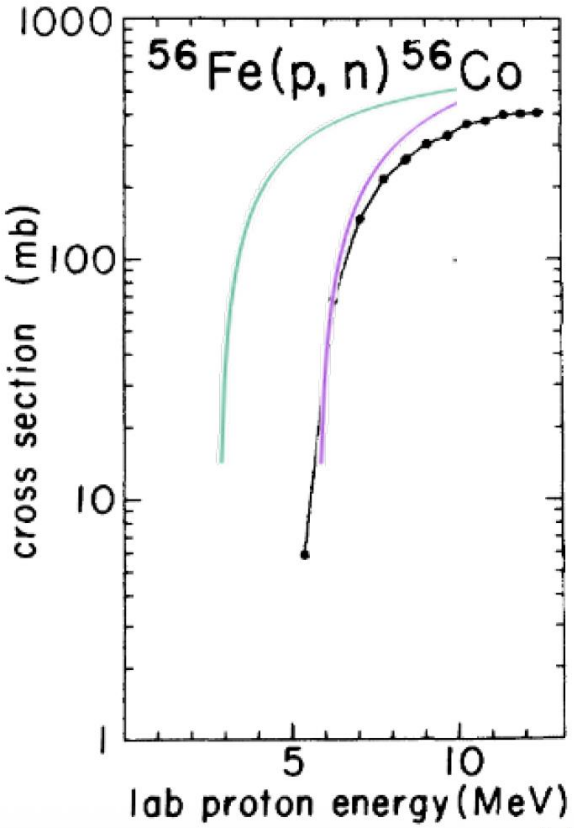
- At closest approach, ($R = R_1 + R_2$), the initial CM energy ε is reduced by $\varepsilon' = \varepsilon - V_C$ so the momentum is $\mathbf{p} = \sqrt{\mathbf{2\mu(\varepsilon - V_c}} = \sqrt{\mathbf{2\mu\varepsilon}}\sqrt{\mathbf{1 - \frac{V_C}{\varepsilon}}}$
- The orbital angular momentum will then be $\ell = \boldsymbol{R\sqrt{2\mu\varepsilon}}\sqrt{\mathbf{1 - \frac{V_C}{\varepsilon}}}$
- Then, $\boldsymbol{\sigma_{total}} = \boldsymbol{\pi}\left(\frac{\boldsymbol{\lambda}}{\mathbf{2\pi}}\right)^{\mathbf{2}}\sum_\ell(\mathbf{2}\ell + \mathbf{1}) = \boldsymbol{\pi}(\ell_{\max} + 1)^2\left(\frac{\boldsymbol{\lambda}}{\mathbf{2\pi}}\right)^{\mathbf{2}} = \boldsymbol{\pi R^2}\left(\mathbf{1} - \frac{\boldsymbol{V_C}}{\boldsymbol{\varepsilon}}\right)$

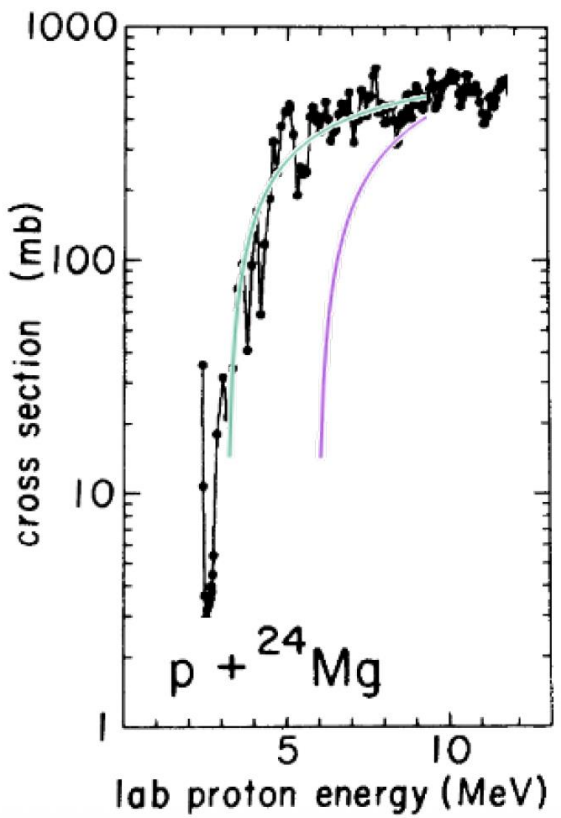
Cross Sections from a Semi-Classical View: With Coulomb

- The orbital angular momentum will then be $\ell = R\sqrt{2\mu\varepsilon}\sqrt{1-\frac{V_C}{\varepsilon}}$

- Then, $\sigma_{total} = \pi\left(\frac{\lambda}{2\pi}\right)^2 \sum_\ell (2\ell+1) = \pi(\ell_{\max}+1)^2\left(\frac{\lambda}{2\pi}\right)^2 = \pi R^2\left(1-\frac{V_C}{\varepsilon}\right)$







P. Dyer et al. PRC 1981

Example Problem

Calculate the total reaction cross section for the ^{48}Ca + ^{208}Pb reaction above the Coulomb barrier

$$\sigma_{total} = \pi R^2 \left(1 - \frac{V_C}{\varepsilon}\right)$$

$$R = R_{Pb} + R_{Ca} = 1.2\text{fm}(208^{1/3} + 48^{1/3}) = 11.47\text{fm}$$

$$V_C = \frac{Z_1 Z_2 e^2}{R} = \frac{80 \times 20 \times 1.22\ MeV - fm}{11.47\ fm} = 205.9\ MeV$$

ϵ (MeV)	σ_{total} (mb)
208	41.7
210	80.7
220	264.9
230	433.1
240	587.2
250	729.1

Terminology

- The total interaction probability is described by the **cross section**
- The cross section as a function of projectile energy is called an **excitation function** (not so commonly used in practice)
- The probability of an interaction ejecting an outgoing particle at a given angle is referred to as the **differential cross section**, written as $\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}$
- The probability of an interaction ejecting an outgoing particle at a given angle with a given energy is referred to as the **double differential cross section**, $\frac{\mathrm{d}^2\sigma}{\mathrm{d}\Omega\mathrm{d}E}$
- The probability of an interaction ejecting an outgoing particle at a given energy is described by the **particle spectrum**

Wrapping up!

- HW #3 due TODAY **at 11:59PM**
- No Office Hours this week again due to DNP Conference
  - Please e-mail me to schedule time to meet
- **Next class**: Scattering
- Questions?

Midterms

- You all did well!
- I want you to learn in this class, so if you want to gain points back from the midterm, you can re-submit to gain points back.
- Average was 87%
- **Midterm re-submissions are due on Oct. 21 in class**